

PERSPECTIVE

AI IN MEDICINE

A Licensure Framework for Autonomous Clinical AI

Alon Bergman, PhD; Robert M. Wachter, MD; Ezekiel J. Emanuel, MD, PhD

The US is nearing a critical juncture in the delivery of clinical care. Physician supply is tightening while demographic trends increase demand. Simultaneously, artificial intelligence (AI) systems, particularly large language models, are beginning to demonstrate competencies that overlap meaningfully with the cognitive tasks integral to clinical care.

The clinician shortage and rapid advances in AI are usually treated as separate policy issues. Both must be considered together. The workforce shortfall creates urgent demand for autonomous clinical AI—agents that make care determinations independently, without per-case clinician review—whereas the adaptive nature of that technology and the risks associated with many clinical decisions demand a fundamentally different regulatory framework. Existing regulatory frameworks for clinical AI are ill suited to adaptive, general-purpose systems. Instead, a licensure-based approach, grounded in ongoing clinical evaluation, offers a safer path forward.

Viewpoint [page 1749](#)**The Workforce Crisis**

National projections point to shortfalls of tens of thousands of physicians during the next decade.¹ These numbers reflect both demographic trends (ie, an aging population and increasing chronic disease burden) and accelerating attrition from clinical practice itself, particularly in primary care. Immigration restrictions threaten to worsen the shortage.² Rural areas are hit hardest; they have lost more than a thousand family physicians in just 6 years, leaving hundreds of counties with minimal or no primary care presence.^{3,4}

Traditional policy responses, such as expanding medical school slots and residency positions, operate on timelines measured in years or decades. Artificial intelligence offers 2 pathways to closing this gap more rapidly. The first is productivity augmentation. Artificial intelligence can extend existing clinician capacity through documentation assistance and clinical decision support. However, such gains are ultimately bounded by clinician supply and the risk that increased throughput may accelerate burnout.⁵ The second is autonomous clinical substitution. Artificial intelligence can provide clinical care without direct clinician oversight. Autonomous AI could markedly improve access. Realizing this latter pathway safely requires a regulatory framework capable of authorizing safe and effective autonomous AI clinical function.

The Entry of AI With Remarkable New Capabilities

During the past few years, AI has undergone a transformation.⁶ Generative AI systems differ fundamentally from earlier clinical AI designed for restricted tasks such as image classification or proto-

col reminders. Autonomous AI technologies operate over natural language, integrate heterogeneous information, and generate reasoned responses in ways akin to clinical cognition.

Within 2 years, generative AI moved from barely passing medical licensing examinations to demonstrating clinical reasoning that expert evaluators found comparable to that of physicians regarding complex cases.^{7,8} Newer models demonstrate even stronger performance on diagnostic tasks, treatment recommendations, and empathetic patient encounters.⁹

Performance on written examinations provides important but incomplete evidence of clinical competence. The critical question is whether AI systems safely improve care in clinical practice with real patients. Recent data suggest they can. A 2025 prospective study of nearly 40 000 primary care visits in Kenya found substantially fewer diagnostic and treatment errors with AI-based decision support.¹⁰ A randomized trial in Pakistan found large improvements in diagnostic reasoning among physicians trained to use large language models effectively.¹¹ The benefits are not automatic. A US trial found no improvement without training or workflow integration, but when implemented thoughtfully, AI can improve clinical decision-making.¹² Most recently, the NOHARM study demonstrated that the “strongest large language model outperformed physicians on safety” and completeness, with clinicians not outperforming large language models on any metric.¹³

These capabilities align closely with the core cognitive work of clinical medicine: reviewing medical records, eliciting patient information, generating differential diagnoses, recommending guideline-concordant interventions, and managing chronic diseases. Complex cases and procedures will continue to require human clinicians. But for a substantial portion of cognitive clinical work—especially in primary care, in which the workforce shortfall is most acute—AI systems can provide clinically adequate support or independent function safely, assuming robust triage to a clinician when needed.⁶

The Inadequacy of Food and Drug Administration–Centered AI Regulation

Other commentators have documented the inadequacy of applying the US Food and Drug Administration (FDA) model for regulation of drugs and devices to generative AI.^{14,15} The FDA’s onetime premarket approval framework fits poorly with adaptive, general-purpose AI that operates across clinical contexts and continuously improves. Recent FDA guidance on predetermined change control plans accommodates some modifications but still requires a onetime certification and a lockdown of the tool until the next recertification.¹⁶ This framework is an advance, but it breaks down for generative AI models that undergo frequent updates across clinical functions.

Table. Comparison of FDA Device Regulation and Licensure for Autonomous Clinical AI

Criterion	FDA device regulation	Licensure framework
Regulatory paradigm	Product approval	Agent credentialing
Initial evaluation	Premarket review based on safety and efficacy for intended use	Standardized examinations and supervised clinical evaluation
Temporal model	Point-in-time clearance with postmarket surveillance	Ongoing, renewable authorization requiring periodic reassessment
Performance monitoring	Adverse event reporting (MDR); voluntary registries	Clinical patient outcomes; peer review; remediation or suspension
Version management	Modifications may require new submissions; predetermined change control plans allow limited prespecification	Competency reevaluated at regular intervals (eg, biennial)
Scope of assessment	Device-specific, task-bounded indications	Clinical reasoning across multiple domains within a defined scope of practice
Locus of accountability	Manufacturer	Developer (primary); deploying institution (secondary)
Response to harm	Recalls, safety communications, labeling changes	Restriction, suspension, or revocation of practice privileges
Regulatory analog	Medical devices, diagnostic software (eg, imaging AI)	Physicians, nurse practitioners, physician assistants

Abbreviations: AI, artificial intelligence; FDA, US Food and Drug Administration; MDR, medical device reporting.

Regulatory Fragmentation Is Accelerating

Compounding this misalignment is growing state-level regulatory fragmentation. In the absence of a comprehensive federal framework for clinical AI, at least 47 states are considering more than 250 clinical AI bills and some have enacted laws addressing data privacy, transparency, bias, and payment policies.¹⁷ For instance, California prohibits health insurers from using AI to deny coverage based on medical necessity ([SB 1120](#)), whereas Colorado mandates bias assessments for high-risk AI systems ([SB 24-205](#)), generating complexity and uncertainty rather than efficient protection of patient safety.

Recent federal action has attempted to address this fragmentation through executive authority. A December 2025 executive order directed federal agencies to preempt state AI regulations but did so without establishing a comprehensive federal framework.¹⁸ This approach has drawn legal challenges and does not resolve the underlying governance gap.

Licensure as a Model for Regulation of Autonomous Clinical AI

Autonomous clinical AI refers to systems that make care determinations, including diagnoses, treatment recommendations, and triage decisions, without clinician review of each case. This method differs from clinical decision support, in which a licensed clinician retains authority over each recommendation. A radiology AI reviewed by a radiologist before clinical action remains decision support, whereas an AI providing primary care consultations without per-case review does not. We propose a framework that applies to autonomous systems only, consistent with FDA's existing decision support exemption.¹⁹

As clinical AI systems become increasingly capable of operating autonomously, a licensing framework may offer a safer and more coherent regulatory approach ([Table](#)). Prior proposals have suggested adapting clinician licensure models to govern clinical AI, emphasizing standardized examinations, supervised clinical use, and periodic reevaluation.^{14,15} We support this approach, extending it in 3 fundamental respects by (1) focusing explicitly on autonomous clinical agents; (2) grounding licensure in workforce constraints that make autonomous clinical AI not merely possible but also necessary; and (3) addressing the regulatory fragmentation already emerging across states. At a minimum, a licensure-like framework for clinical AI should include 6 elements.

First, competency certification could begin with standardized examinations. Each autonomous AI clinical model would be required to perform at or above the median score of most recent human test takers on all 3 components of the US Medical Licensing Examination and any relevant specialty board examinations aligned with its intended scope. Passing scores would establish minimum competency, not clinical readiness.

Second, models that meet this threshold would enter a supervised clinical deployment period analogous to residency training and requiring demonstration of noninferior clinical performance. Specific requirements for patient volume would be established through the multistakeholder standards process.

Third, this licensure-like approach would define a scope of practice, specifying which clinical functions specific AIs may perform, in which settings, and under what level of oversight. For example, an AI certified for primary care triage could gather histories and recommend next steps but could not prescribe care or medications without clinician approval.

Fourth, certification would be time limited and contingent on ongoing performance monitoring and reevaluation. Because AI systems are adaptive rather than static, continued authorization would depend on periodic (eg, biennial) demonstration of acceptable levels of clinical performance.

Fifth, accountability structures must be explicit. Developers bear primary responsibility for model performance, including safety profile, accuracy, and behavior across clinical contexts. Deploying institutions are responsible for implementation: integrating the AI into clinical workflows, ensuring that the conditions covered under the AI approval (including triaging to a clinician when appropriate) are followed, monitoring patient outcomes, and reporting adverse events. This layered structure mirrors existing liability frameworks in medicine.

Sixth, federal preemption would be necessary to avoid 50 parallel licensing regimens. Under this approach, federal certification of AI competency would be binding on all states, whereas states would retain authority over scope of practice, supervision requirements, and enforcement.

Recognizing autonomous AI's rapid evolution, this licensure-like oversight would subject autonomous clinical AI models to more continuous, comprehensive, flexible, and clinical evaluation than the FDA's current Software as a Medical Device framework ([Table](#)).

To illustrate, an AI providing autonomous primary care consultations in rural federally qualified health centers would first pass standardized competency examinations and then enter supervised deployment, with cases periodically reviewed and performance tracked, before receiving unrestricted licensure for a defined scope of practice, subject to biennial renewal. The developer would be

accountable for model performance; the federally qualified health center leadership, including its clinicians, would be responsible for workflow integration, supervision protocols, and adverse event reporting. States may impose additional practice restrictions based on local needs. Although primary care represents the most urgent need, this framework applies across specialty domains; the more confined repertoires of many subspecialties may simplify safety and competency assessments.

Federal Governance Architecture

Implementing this licensure framework requires new institutional infrastructure. We propose establishing an Office of Clinical AI Oversight within the US Department of Health and Human Services, staffed by individuals with dual expertise in clinical medicine and AI (practicing physicians, clinical informaticians, machine learning engineers, and health policy experts) and an external advisory board representing medical specialty societies, patient advocacy groups, AI developers, and academia.

Standards for autonomous clinical AI licensure would emerge through a multistakeholder consensus process, with working groups developing specialty-specific competency frameworks, performance benchmarks, and certification protocols through iterative, transparent revision. This process would draw on lessons from broad clinical accreditors, such as the Accreditation Council for Graduate Medical Education, the Joint Commission, and the specialty certifying boards.

Federal-State Coordination

Federal preemption is necessary to establish uniform competency standards and avoid fragmented certification across 50 states while preserving state authority over practice conditions and enforcement.

Here, the appropriate model is the American Board of Medical Specialties' board certification process. States cannot require physicians to redemonstrate competency beyond what the board certifies. For AI licensure, federal law would similarly establish that states must recognize AI systems certified by the federal Office of Clinical AI Oversight. States would retain authority over scope of practice, supervision requirements, and enforcement but could not require separate competency assessments beyond the federal standard.

Federal preemption of state AI laws requires congressional legislation. Executive action alone cannot establish the supreme law of

the land under the Constitution's Supremacy Clause.¹⁸ Our proposal, therefore, envisions statutory authorization for the Office of Clinical AI Oversight, not executive fiat.

Clear developer pathways, including transparent timelines and presubmission consultations, would balance rigorous oversight with innovation.¹⁷

Objections Considered

Three objections merit consideration. First, medical licensure has traditionally been a state function. However, like telemedicine, autonomous clinical AI systems will be deployed across state lines, making fragmented, state-by-state oversight inefficient and poorly aligned with how these technologies operate. A nationally coordinated licensure framework would better reflect cross-state deployment and reduce the sludge of duplicative approvals and administrative costs.

Second, licensure might imply equivalence between autonomous clinical AI systems and physicians. Clinicians are likely to remain integral for complex judgment, moral reasoning, and patient counseling. Regulatory standards and decisions for clinical AI need to be explicit about when oversight is required and equally explicit about when autonomous clinical AI can be safely deployed.

Third, implementation capacity is a concern. The US Department of Health and Human Services currently lacks the infrastructure to evaluate autonomous AI in health care at scale, and performance benchmarks have not yet been established. Funding through developer fees, analogous to FDA user fees, could support the necessary infrastructure.

Conclusions

Rapidly advancing autonomous clinical AI presents a regulatory challenge that the FDA's current framework, built for drugs and devices, poorly addresses. That framework assumes static products, narrow indications, and clear manufacturer accountability, assumptions that autonomous clinical AI defies.

Licensure is better aligned with the realities of autonomous clinical AI. It will not resolve broader issues related to data governance, reimbursement, privacy, workforce transformation, or the effect of local deployment, all of which may require additional regulation. As clinical AI increasingly resembles clinicians in its capabilities, our regulatory frameworks must evolve accordingly.

ARTICLE INFORMATION

Author Affiliations: Department of Medical Ethics and Health Policy, University of Pennsylvania, Philadelphia (Bergman, Emanuel); Department of Medicine, University of California, San Francisco (Wachter).

Corresponding Author: Alon Bergman, PhD, Department of Medical Ethics and Health Policy, University of Pennsylvania, 423 Guardian Dr, Philadelphia, PA 19104 (alon.bergman@pennmedicine.upenn.edu).

Published Online: April 29, 2026.
doi:10.1001/jama.2026.5483

Conflict of Interest Disclosures: Dr Wachter reported royalties from Portfolio/Penguin and stock options from Arbitrator, Curai, Notable, and Commure outside the submitted work. Dr Emanuel reported honoraria from Employer Direct Healthcare,

University of California Los Angeles, Baylor College of Medicine, Cedar CEO Council, Koo Foundation, Brown University Emergency Medicine, 10th Annual CVS Accountable Care Symposium, Building Together Healthcare Summit, California Orthopaedic Association, HCRX, Brocher Summer Academy, RAISE Symposium, Spring Tide, Avalon Health Care Solutions, Massachusetts Association of Health Plans, UCSF Department of Medicine, and United Health Care Group; travel reimbursement from the TUM Institute for History and Ethics of Medicine, BCEPS International Symposium, New York Historical Society, and UVA Darden School of Business outside the submitted work; lodging reimbursement from Arendalsuka and Amangiri; serving on the advisory board for Daymark Health, Peterson Center on Healthcare, JSL Health Capital, Cellaes, Clarify Health Solutions, and FeelBetter; consulting for Korro/

Coach AI; and serving as an advisor for Link Health Technology, Healthcare Foundry Inc, Dendro Technologies, Notable Health, and Smirk Health. No other disclosures were reported.

REFERENCES

1. Association of American Medical Colleges. The complexities of physician supply and demand: projections from 2021 to 2026. Published March 2024. Accessed February 2, 2026. <https://www.aamc.org/media/75231/download>
2. Rotenstein LS, He Z, Dziura J, et al. Trends in and predictors of physician attrition from clinical practice across specialties: a nationwide, longitudinal analysis. *Ann Intern Med*. 2025;178(12):1698-1708. doi:10.7326/ANNALS-25-00564
3. Horstman C, Shah A. The state of rural primary care in the United States. Commonwealth Fund.

Published November 17, 2025. Accessed February 13, 2026. <https://www.commonwealthfund.org/publications/issue-briefs/2025/nov/state-rural-primary-care-united-states>

4. Fogarty CT, Byun H, Huffstetler AN. Family physician workforce trends: the toll on rural communities. *Ann Fam Med*. 2025;23(6):535-538. doi:10.1370/afm.240549
5. Liu H, Ding N, Li X, et al. Artificial intelligence and radiologist burnout. *JAMA Netw Open*. 2024;7(11):e2448714. doi:10.1001/jamanetworkopen.2024.48714
6. Wachter RM. *A Giant Leap: How AI Is Transforming Healthcare and What That Means for Our Future*. Penguin Publishing Group; 2026.
7. Kung TH, Cheatham M, Medenilla A, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. *PLOS Digit Health*. 2023;2(2):e0000198. doi:10.1371/journal.pdig.0000198
8. Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. *Nature*. 2023; 620(7972):172-180. doi:10.1038/s41586-023-06291-2
9. McDuff D, Schaeckermann M, Tu T, et al. Towards accurate differential diagnosis with large language models. *Nature*. 2025;642(8067):451-457. doi:10.1038/s41586-025-08869-4
10. Korom R, Kiptinness S, Adan N, et al. AI-based clinical decision support for primary care: a real-world study. *arXiv*. Preprint posted online July 22, 2025. doi:10.48550/arXiv.2507.16947
11. Qazi IA, Ali A, Khawaja AU, Akhtar MJ, Sheikh AZ, Alizai MH. Large language model diagnostic assistance for physicians in a lower-middle-income country: a randomized controlled trial. *Nature Health*. 2026;1(2):198-205. doi:10.1038/s44360-025-00007-8
12. Goh E, Gallo R, Hom J, et al. Large language model influence on diagnostic reasoning: a randomized clinical trial. *JAMA Netw Open*. 2024; 7(10):e2440969. doi:10.1001/jamanetworkopen.2024.40969
13. Wu D, Haredasht FN, Maharaj SK, et al. First, do NOHARM: towards clinically safe large language models. *arXiv*. Preprint posted online December 1, 2025. doi:10.48550/arXiv.2512.01241
14. Bressman E, Shachar C, Stern AD, Mehrotra A. Software as a medical practitioner—is it time to license artificial intelligence? *JAMA Intern Med*. 2026;186(1):5-6. doi:10.1001/jamainternmed.2025.6132
15. Blumenthal D, Patel B. The regulation of clinical artificial intelligence. *NEJM AI*. Published July 12, 2024. Accessed January 18, 2026. <https://ai.nejm.org/doi/full/10.1056/AIpc2400545>
16. US Food and Drug Administration. Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions: guidance for industry and Food and Drug Administration staff. Published December 4, 2024. Updated August 18, 2025. Accessed March 30, 2026. <https://www.fda.gov/media/166704/download>
17. Blumenthal D, Marellapudi A. More fragmented, more complex: state regulation of AI in health care. *NEJM AI*. Published May 5, 2025. Accessed January 18, 2026. <https://ai.nejm.org/doi/full/10.1056/AIpc2500163>
18. Shachar C, Blumenthal D, Cohen IG, Huberfeld N. Preemption at the intersection of health care and artificial intelligence. *JAMA*. 2026;335(10):845-846. doi:10.1001/jama.2026.0061
19. Karnik K. FDA regulation of clinical decision support software. *J Law Biosci*. 2014;1(2):202-208. doi:10.1093/jlb/lisu004